exponent → b^x ← base "b to the power of x" or "b to the x" 5^3 = "5 to the power of 3"

A.4: Exponent Rules Toolkit

Term	Explanation	Examples
Base b	what you are multiplying repeatedly	$2^3, 2^5, 2^{10}, 2^{200}$ ← base is 2
Exponent x	how many times you multiply the base	$2^7, 5^7, 60^7$ ← exponent is 7
Expanded Form (or Standard Form)	show each multiplication	$3 \cdot 3 \cdot 3 \cdot 3$
Exponential Form	simplifying using exponents	3^4

Exponent Rule	Written Explanation and Example	Why it Works
Multiplicative (Product) Law (Multiplying Terms) $x^n \cdot x^m = x^{n+m}$	*When you multiply terms with the same base…* you can add the exponents together $2^3 \cdot 2^5 = 2^{3+5} = 2^8$ $5^2 \cdot 5^7 = 5^{2+7} = 5^9$	$5^2 \cdot 5^7 = (5 \cdot 5) \cdot (5 \cdot 5 \cdot 5 \cdot 5 \cdot 5 \cdot 5 \cdot 5)$ $= 5 \cdot 5 \cdot 5 \cdot 5 \cdot 5 \cdot 5 \cdot 5 \cdot 5 \cdot 5$ $= 5^9$
Quotient Law (Dividing Terms) $\frac{x^n}{x^m} = x^{n-m}$	*When you divide terms with the same base…* you can subtract the expo $\frac{2^5}{2^3} = 2^{5-3} = 2^2$ $\frac{7^5}{7^1} = 7^{5-1} = 7^4$	$\frac{2^5}{2^3} = \frac{2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 \cdot 1}{2 \cdot 2 \cdot 2 \cdot 1}$ $= \frac{2^2}{1}$ $= 2^2$
Power Law (Taking the power of a power) $(x^n)^m = x^{n \cdot m}$	*When you raise a whole term to a power..* you can multiply the exponents together $(2^5)^4 = 2^{5 \cdot 4} = 2^{20}$ $(3^6)^5 = 3^{6 \cdot 5} = 3^{30}$	$(5^2)^3 = 5^2 \cdot 5^2 \cdot 5^2$ $= 5^{2+2+2} = 5^{2 \cdot 3}$ $= 5^6$
Zero Exponents $x^0 = 1$	*When a base or term is raised to a zero exponent…* the value will always be one. $5^0 = 1$ $100^0 = 1$ $1{,}586{,}310^0 = 1$	$\frac{2^4}{2^4} = \frac{2 \cdot 2 \cdot 2 \cdot 2 \cdot 1}{2 \cdot 2 \cdot 2 \cdot 2 \cdot 1} = \frac{1}{1} = 1$ ↳ expect $= 2^{4-4} = 2^0$
Negative Exponents $x^{-n} = \frac{1}{x^n}$	*When a base or term is raised to a negative exponent…* you flip into a fraction and change the exponent to positive $3^{-2} = \frac{1}{3^2}$ $100^{-99} = \frac{1}{100^{99}}$ $\left(\frac{1}{2}\right)^{-3} = \left(\frac{2}{1}\right)^3 = 2^3$	$\frac{2^3}{2^5} = \frac{2 \cdot 2 \cdot 2 \cdot 1}{2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 \cdot 1} = \frac{1}{2^2}$ ↳ expect $2^{3-5} = 2^{-2}$
Fractional Exponents 1 $x^{\frac{1}{q}} = \sqrt[q]{n}$	*When a base or term is raised to an exponent that is a fraction with 1 in the numerator…* we can rewrite to → the "denominator" root of the base. $2^{\frac{1}{3}} = \sqrt[3]{2}$ $7^{\frac{1}{50}} = \sqrt[50]{7}$ $5^{\frac{1}{6}} = \sqrt[6]{5}$	$2^1 = 2^{\frac{3}{3}} = 2^{\frac{1}{3}+\frac{1}{3}+\frac{1}{3}}$ $= 2^{\frac{1}{3}} \cdot 2^{\frac{1}{3}} \cdot 2^{\frac{1}{3}}$ $= \sqrt[3]{2} \cdot \sqrt[3]{2} \cdot \sqrt[3]{2}$
Fractional Exponents 2 $x^{\frac{p}{q}} = (\sqrt[q]{n})^p$ or $\sqrt[q]{n^p}$	*When a base or term is raised to an exponent that is a fraction…* we can rewrite to → the "denominator" root to the power of "numerator" $2^{\frac{4}{3}} = (\sqrt[3]{2})^4 = \sqrt[3]{2^4}$ $5^{\frac{2}{7}} = (\sqrt[7]{5})^2 = \sqrt[7]{5^2}$	$2^{\frac{4}{3}} = 2^{\frac{1}{3} \cdot 4} = (2^{\frac{1}{3}})^4 = (\sqrt[3]{2})^4$ or $2^{4 \cdot \frac{1}{3}} = (2^4)^{\frac{1}{3}} = \sqrt[3]{2^4}$
Same Exponents $x^n \cdot y^n = (xy)^n$	*When two base terms are multiplied together with the same exponent…* you can multiply the bases and retain (keep) the exponent $3^2 \cdot 4^2 = (3 \cdot 4)^2 = 12^2$ $10^7 \cdot 3^7 = (10 \cdot 3)^7 = 30^7$	$5^3 \cdot 2^3 = (5 \cdot 5 \cdot 5) \cdot (2 \cdot 2 \cdot 2)$ $= 5 \cdot 2 \cdot 5 \cdot 2 \cdot 5 \cdot 2$ $= (5 \cdot 2)^3$